

Practicum No.8**Date: 20/2/2026****Loan/Grant Application Preparation for a CSE/AIML/DSCS/DS Start-up****I. Practical Significance**

Preparing a loan or grant application for a CSE, AIML, DSCS, or DS-based startup holds significant practical value because it strengthens the venture's ability to secure essential funding for product development, cloud infrastructure, research, and market deployment. The process exposes students and founders to real-world requirements such as articulating a compelling problem statement, defining a technically sound solution, estimating computational resource needs, and presenting measurable outcomes using AI/ML, data science, and cloud technologies. It builds competency in preparing budgets for training AI models, deploying cloud-native systems, developing analytics pipelines, and ensuring data security compliance. In addition, the preparation process reinforces understanding of incubator support mechanisms, TRL-based project planning, risk assessment, and commercialization strategies. By drafting a formal proposal, students gain hands-on experience in aligning technological innovation with financial feasibility, societal impact, and investor expectations. Ultimately, this activity equips emerging technopreneurs with the skills to pursue government grants, innovation fellowships, accelerator funding, and soft loans—critical enablers for scaling high-tech ventures in competitive computing domains.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a pitch deck, project proposals for small/medium scale startups logically and scientifically.
 - a) **Interpret** competitive dynamics shaping CSE-focused high-tech and software-driven markets.(CDO)
 - b) **Evaluate** commercial viability of CSE products such as AI models, software applications, analytics platforms, and cloud-native solutions
 - c) **Innovate** strategic direction for new ventures in advanced CSE domains including generative AI, distributed systems, and intelligent automation.
 - a) **Practice** professional ethics in data collection, system design, and user interaction within digital ecosystems.(ADO)

III. Relevant CO(s)

- **CO3: Formulate** a startup proposal for the chosen idea through entrepreneurial support systems.

IV. Practical Outcome (PrO)

PrO8: Present a filled-out loan application form or grant application relevant to **the identified** CSE, AIML, DSCS & DS start-up.

V. Related ADO(s) (keep as such)

- a) **Practice** professional ethics.
- b) **Function** effectively as a team member/leader.
- c) **Practice** integrity and honesty.

VI. Minimum Underpinning Theory

The theoretical foundation for this practicum is rooted in core principles of computing entrepreneurship, technology-driven innovation, and ethical digital system design. Fundamental theories such as software engineering life cycles, cloud-native architectures, AI/ML model development workflows, and data analytics pipelines provide the basis for understanding how CSE products are engineered and deployed at

scale. Market-oriented theories including Porter's Five Forces, technology diffusion models, and competitive advantage frameworks explain how high-tech ventures operate within dynamic and rapidly evolving software-driven markets. The practicum also draws on innovation management theories such as Lean Startup, Design Thinking, and TRL (Technology Readiness Levels), which help analyze the strategic direction and commercialization pathways available to startups in advanced computing domains like generative AI and distributed systems. Additionally, ethical frameworks including Responsible AI, privacy-by-design, cybersecurity principles, and user-centric design ethics underpin the responsible development of digital products and platforms. Together, this theoretical grounding enables students to interpret competition, assess commercial viability, envision strategic venture opportunities, and uphold ethical practices during their engagement with the entrepreneurship support agency or incubator.

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and documentation	1 per team
2	Internet access	For scheme research	As required
3	Funding templates	Govt/Incubator formats	1 set
4	Reference materials	Scheme PDFs, institutional portals	1 per team

IX. Safety Precautions

- Ensure** data accuracy and avoid falsified figures.
- Ensure** confidentiality of organizational data.
- Maintain** ethical and transparent financial reporting.
- Respect** compliance norms of financial institutions.

X. Procedure

- Form a team and select a domain (CSE/AIML/DSCS/DS).
- Research relevant loan/grant schemes (MSME, Startup India, DST, MEITY, etc.).
- Identify the funding needs of the selected startup idea.
- Prepare:
 - Executive summary
 - Technical justification
 - Market need
 - Financial projections
- Draft the loan/grant application using official templates.
- Review and refine the proposal.
- Submit to faculty for evaluation.

XI. Actual Resources Used *(To be filled by the students)*

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	ASUS	Used for research, proposal preparation, and documentation	1	N/A

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
2.	Internet Access	Wi-Fi	Used to search funding schemes and application templates	1	N/A
3.	Reference Materials	Govt Websites	Startup India, MSME, DST scheme information	1	N/A
4.	Funding Templates	Online Templates	Loan/grant application format and business plan template	1	N/A

XII. Actual Procedure Followed

1. First, we selected **Digital Attendance and Productivity Tracker** as the startup idea for preparing the loan/grant application.
2. Then we identified the **funding requirements** needed for developing the system, such as software development, cloud services, database hosting, and testing tools.
3. After that, we researched **government funding schemes such as Startup India Seed Fund Scheme and MSME Startup Support Programs**.
4. A suitable scheme was selected based on its support for **technology-based startup ideas and early-stage innovation**.
5. We used an **online business proposal template to prepare the startup funding application**, including sections such as executive summary, problem statement, and proposed solution.
6. The proposal also included **technical details, target users, market need, and estimated financial requirements**.

XIII. Observations and Data

S. No.	Loan /Grant application identified	Data/ Remarks
1	Startup India Seed Fund Scheme	Supports early-stage startups with funding and mentorship
2	MSME Startup Scheme	Provides financial assistance and business development support
3	Funding Requirement	Software development, cloud hosting, and system deployment
4	Target Users	Educational institutions, teachers, and students
5	Expected Outcome	Development of a digital platform for attendance tracking and productivity monitoring

XIV. Results/Observations

1. The research showed that several **government funding schemes support innovative technology startups in their early stages**.
2. A well-prepared proposal with **clear technical explanation and financial planning increases the chances of obtaining funding**.
3. Funding can help in **developing the prototype, testing the system, and deploying the Digital Attendance and Productivity Tracker in educational institutions**.

XV. Interpretation of Results

1. The analysis indicates that funding support is important for **developing and scaling technology-based startup ideas**.
2. For the **Digital Attendance and Productivity Tracker**, financial support can help in building the application, maintaining servers, and improving system features.
3. Proper financial planning and proposal preparation increase the chances of **attracting government or institutional funding**.

XVI. Conclusions

1. This practicum helped in understanding the **process of preparing loan and grant applications for technology startups**.
2. It highlighted the importance of **clear proposal writing, realistic budgeting, and strong technical justification**.
3. With proper planning and funding support, the **Digital Attendance and Productivity Tracker** can become a feasible and useful solution for educational institutions.

XVII. Suggested Practicum Related Questions

1. **Evaluate** how financial planning and cost estimation influence the approval of loan/grant proposals for CSE/AIML/DSCS/DS startups.
2. **Assess** how technology readiness level (TRL) impacts eligibility for government and institutional funding schemes.
3. **Examine** the influence of compliance, data privacy, and cybersecurity requirements on funding decisions.
4. **Evaluate** how risk-mitigation strategies strengthen the feasibility of startup loan/grant applications.
5. **Interpret** how funding insights can guide MVP development, operational planning, and long-term financial sustainability.

XVIII References / Suggestions for further reading (keep as such)

- a) Venture Meets Mission: Aligning People, Purpose, and Profit to Innovate and Transform Society, Arun Gupta, Gerard George, Thomas J. Fewer, Stanford University Press, Stanford, California, 2024, ISBN: 978-1503636283
- b) So You Want to Start a Business: Eight Steps to Startup Success, John B. Vinturella, Business Expert Press, New York, USA, 2024, ISBN: 978-1637425855
- c) Business Planning, Franklin A. Gevurtz, Foundation Press (West Academic), St. Paul, Minnesota, 2024, ISBN: 978-1685612764

XIX Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the **process-related skills** and **product-related skills** will change depending on the Pro, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Identification of domain & problems	3	
2	Use of structured idea generation techniques	4	
3	Team collaboration & creativity	4	
4	Feasibility evaluation	3	

Performance Indicators		Maximum Marks	Marks Obtained
5	Ethical originality & clarity	4	
Product-related Skills: 12 Marks - 40%			
1	Quality & novelty of ideas	2	
2	Logical presentation & organization	2	
3	Clarity of findings & conclusions	3	
4	Report/presentation submission	3	
5	Innovation potential & market relevance	2	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Signature with Date
Process-Related skills (18)	Product-Related skills (12)	Total (30)		

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